

RESULTS

Results

01 · MANOVA

groups compared on several outcomes at once

Table 1. Multivariate tests

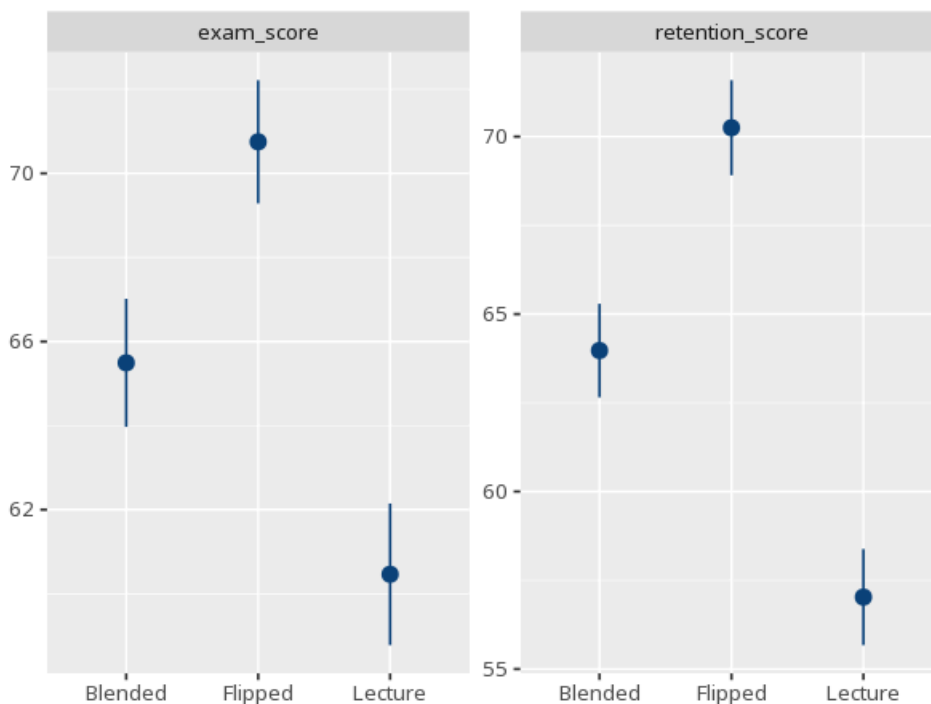
Effect	Pillai / Wilks	approx F	df1	df2	p	partial η^2 [95% CI]
teaching_method	0.46	35.03	4	474	<.001	0.23 [0.17, 1.00]

assumption check: homogeneity of covariance matrices (Box's M). (Box's M $\chi^2=3.11$, df=6, p=.795) — covariance matrices look homogeneous

Table 2. Follow-up univariate ANOVAs (per DV)

DV	F	df1	df2	p	partial η^2 [95% CI]
exam_score	43.07	2	237	<.001	0.27 [0.19, 1.00]
retention_score	97.01	2	237	<.001	0.45 [0.38, 1.00]

Figure. Group means per outcome



Understanding the numbers

Effect. Names which factor (or interaction) the multivariate test is evaluating. Here, the effect tested is teaching_method.

Pillai's V / Wilks' Λ . The chosen multivariate test statistic summarizing how much the groups differ across all outcomes jointly (Pillai's trace is the more robust default). Here, Pillai's $V = 0.46$.

approx F. An F-value approximating the multivariate test statistic's significance; the follow-up table reports the same kind of F computed per single outcome. Here, $\text{approx } F(4, 474) = 35.03$.

df1. The numerator degrees of freedom for the multivariate (or follow-up) test. Here, $df1 = 4$.

df2. The denominator degrees of freedom for the multivariate (or follow-up) test. Here, $df2 = 474$.

p. The probability of a multivariate difference this large if the groups truly had identical outcome profiles. Here, $p < .001$ for the multivariate test - below the $\alpha = 0.05$ threshold.

partial η^2 (multivariate). The effect size for the multivariate test itself, converted directly from its approx F and degrees of freedom (there is no per-cell sum-of-squares breakdown for a multivariate statistic). Here, $\text{partial } \eta^2 = .23 [.17, 1.00]$ for the multivariate test.

DV. Names which single outcome that follow-up row's univariate ANOVA tests. Here, the follow-up table above reports one row for each of the 2 outcomes.

partial η^2 . The proportion of that single outcome's variance explained by the factor, in its follow-up univariate ANOVA - correct these follow-up p-values for multiple comparisons before calling any significant. Here, each of the 2 follow-up ANOVAs above reports its own partial η^2 for that outcome.

How to read this test

Tests whether groups differ on a set of outcomes jointly. Read the multivariate p (Pillai's trace is robust) first; if significant, the per-DV follow-up ANOVAs show which individual outcomes drive it. Because you run one ANOVA per outcome, correct those follow-up p-values for multiple comparisons (e.g. Bonferroni: divide alpha by the number of outcomes) before calling each significant. Your significance threshold (α) is 0.05.

APA template: A MANOVA gave Pillai's $V = .46$, $F(4, 474) = 35.03$, $p < .001$, $\text{partial } \eta^2 = .23 [.17, 1.00]$.

Statistical basis: Pillai's trace multivariate test (headline statistic) (Pillai, K. C. S. (1955). "Some new test criteria in multivariate analysis." *Annals of Mathematical Statistics*, 26(1), 117-121.); Wilks' Lambda multivariate test (Wilks, S. S. (1932). "Certain generalized distributions in multivariate analysis." *Biometrika*, 24(3/4), 471-494.); Box's M test of homogeneity of covariance matrices (Box, G. E. P. (1949). "A general distribution theory for a class of likelihood criteria." *Biometrika*, 36(3-4), 317-346.); Multivariate effect size (partial η^2 from the multivariate approx. F) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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